CLASS 10 · CBSE · NCERT MATHEMATICS · SHORT NOTES

Chapter 8: Introduction to Trigonometry

Topic-wise short notes with solved examples

Trigonometry relates the sides of a right-angled triangle to its acute angles. This chapter covers the six ratios, their values for standard angles and the three basic identities.

1 Trigonometric ratios

- In a right-angled triangle ABC, right-angled at B, take the acute angle A. Then BC is the side **opposite** to A, AB is the side **adjacent** to A and AC is the **hypotenuse**.

Ratio	Definition	Ratio	Definition
$\sin A$	$\frac{\text{opposite}}{\text{hypotenuse}}$	$\operatorname{cosec} A$	$\frac{\text{hypotenuse}}{\text{opposite}}$
$\cos A$	$\frac{\text{adjacent}}{\text{hypotenuse}}$	$\sec A$	$\frac{\text{hypotenuse}}{\text{adjacent}}$
$\tan A$	$\frac{\text{opposite}}{\text{adjacent}}$	$\cot A$	$\frac{\text{adjacent}}{\text{opposite}}$

- **Reciprocal relations**: $\operatorname{cosec} A = \frac{1}{\sin A}$, $\sec A = \frac{1}{\cos A}$, $\cot A = \frac{1}{\tan A}$.
- **Quotient relations**: $\tan A = \frac{\sin A}{\cos A}$ and $\cot A = \frac{\cos A}{\sin A}$.
- The value of a ratio depends only on the angle, not on the size of the triangle.
- $\sin A$ is one symbol. It does not mean $\sin \times A$.

REMEMBER

The hypotenuse is the longest side, so $\sin A$ and $\cos A$ are never greater than 1, and $\sec A$ and $\operatorname{cosec} A$ are never less than 1.

2 Finding all ratios from one

- Step 1: draw a right-angled triangle and mark two sides using the given ratio, as k times the numbers.
- Step 2: find the third side with the Pythagoras theorem.
- Step 3: write the required ratios.

Example 2.1 If $\sin A = \frac{5}{13}$, find $\cos A$ and $\tan A$.

SOLUTION

Take opposite side $= 5k$ and hypotenuse $= 13k$.

Adjacent side $= \sqrt{(13k)^2 - (5k)^2} = \sqrt{144k^2} = 12k$

$\cos A = \frac{12}{13}$ **and** $\tan A = \frac{5}{12}$

Example 2.2 If $\tan\theta = \frac{3}{4}$, evaluate $\frac{1-\tan^2\theta}{1+\tan^2\theta}$ and show that it equals $\cos^2\theta - \sin^2\theta$.

SOLUTION

$\frac{1-\frac{9}{16}}{1+\frac{9}{16}} = \frac{7/16}{25/16} = \frac{7}{25}$

Opposite $= 3k$, adjacent $= 4k$, hypotenuse $= 5k$, so $\sin\theta = \frac{3}{5}$ and $\cos\theta = \frac{4}{5}$

$\cos^2\theta - \sin^2\theta = \frac{16}{25} - \frac{9}{25} = \frac{7}{25}$

Both sides equal $\frac{7}{25}$.

3 Ratios of standard angles

Angle	$0°$	$30°$	$45°$	$60°$	$90°$
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not defined
cosec	Not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
sec	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not defined
cot	Not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

- To remember the sin row: $\sqrt{\frac{0}{4}}, \sqrt{\frac{1}{4}}, \sqrt{\frac{2}{4}}, \sqrt{\frac{3}{4}}, \sqrt{\frac{4}{4}}$. The cos row is the sin row in reverse.
- As the angle increases from $0°$ to $90°$, sin increases from 0 to 1 and cos decreases from 1 to 0.

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Example 3.1 Evaluate $4\sin^2 60^\circ + 3\tan^2 30^\circ - 8\sin 45^\circ \cos 45^\circ$.

SOLUTION

$= 4 \times \frac{3}{4} + 3 \times \frac{1}{3} - 8 \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}$

$= 3 + 1 - 4$

0

Example 3.2 If $\sin(A+B) = 1$ and $\cos(A-B) = \frac{\sqrt{3}}{2}$, where $0^\circ < A + B \leq 90^\circ$ and $A > B$, find A and B.

SOLUTION

$\sin(A+B) = 1 = \sin 90^\circ$, so $A + B = 90^\circ$

$\cos(A-B) = \frac{\sqrt{3}}{2} = \cos 30^\circ$, so $A - B = 30^\circ$

Adding: $2A = 120^\circ$

$A = 60^\circ$ **and** $B = 30^\circ$

Example 3.3 In $\triangle ABC$, right-angled at B, $AC = 12$ cm and $\angle A = 60^\circ$. Find AB and BC.

SOLUTION

$\cos 60^\circ = \frac{AB}{AC}$, so $AB = 12 \times \frac{1}{2} = 6$ cm

$\sin 60^\circ = \frac{BC}{AC}$, so $BC = 12 \times \frac{\sqrt{3}}{2} = 6\sqrt{3}$ cm

$AB = 6$ **cm and** $BC = 6\sqrt{3}$ **cm**

REMEMBER

$\sin(A+B)$ is not $\sin A + \sin B$. Check with $A = B = 30^\circ$: $\sin 60^\circ = \frac{\sqrt{3}}{2}$, but $\sin 30^\circ + \sin 30^\circ = 1$.

4 Trigonometric identities

- An identity is true for every value of the angle for which both sides are defined.

$$\sin^2 A + \cos^2 A = 1 \qquad 1 + \tan^2 A = \sec^2 A \qquad 1 + \cot^2 A = \operatorname{cosec}^2 A$$

- Each identity can be rearranged: $\sin^2 A = 1 - \cos^2 A$, $\sec^2 A - \tan^2 A = 1$, $\operatorname{cosec}^2 A - \cot^2 A = 1$.
- $\sin^2 A$ means $(\sin A)^2$.

Example 4.1 Express $\cos A$ and $\tan A$ in terms of $\sin A$.

SOLUTION

From $\sin^2 A + \cos^2 A = 1$: $\cos A = \sqrt{1 - \sin^2 A}$

$$\tan A = \frac{\sin A}{\cos A} = \frac{\sin A}{\sqrt{1 - \sin^2 A}}$$

$\cos A = \sqrt{1 - \sin^2 A}$ **and** $\tan A = \dfrac{\sin A}{\sqrt{1 - \sin^2 A}}$

Example 4.2 Prove that $(1 + \tan^2 A)(1 - \sin A)(1 + \sin A) = 1$.

SOLUTION

$$\text{LHS} = \sec^2 A\,(1 - \sin^2 A)$$

$$= \sec^2 A \times \cos^2 A = \frac{1}{\cos^2 A} \times \cos^2 A$$

$= 1 =$ **RHS**

Example 4.3 Prove that $\tan A + \cot A = \sec A \operatorname{cosec} A$.

SOLUTION

$$\text{LHS} = \frac{\sin A}{\cos A} + \frac{\cos A}{\sin A} = \frac{\sin^2 A + \cos^2 A}{\sin A \cos A}$$

$$= \frac{1}{\sin A \cos A} = \frac{1}{\cos A} \times \frac{1}{\sin A}$$

$= \sec A \operatorname{cosec} A =$ **RHS**

Example 4.4 Prove that $\sqrt{\dfrac{1 - \cos A}{1 + \cos A}} = \operatorname{cosec} A - \cot A$.

SOLUTION

Multiply the numerator and denominator inside the root by $(1 - \cos A)$:

$$\text{LHS} = \sqrt{\frac{(1 - \cos A)^2}{1 - \cos^2 A}} = \sqrt{\frac{(1 - \cos A)^2}{\sin^2 A}} = \frac{1 - \cos A}{\sin A}$$

$$= \frac{1}{\sin A} - \frac{\cos A}{\sin A}$$

$= \operatorname{cosec} A - \cot A =$ **RHS**

5 How to prove an identity

- Start from the more complicated side and work towards the simpler one.
- If nothing else suggests itself, change every ratio to $\sin$ and $\cos$.
- For a sum of fractions, take the LCM. For $1 \pm \sin A$ or $1 \pm \cos A$ in a denominator, multiply by the conjugate.
- Look for $1 - \sin^2 A$, $1 - \cos^2 A$, $\sec^2 A - 1$ and $\operatorname{cosec}^2 A - 1$: each one is a single square.
- Never move terms from one side to the other in a proof. Work on one side only, or simplify both sides separately.

6 Common mistakes

- Writing $\sin A = \dfrac{\text{opposite}}{\text{adjacent}}$. That is $\tan A$.
- Treating $\sin^2 A$ as $\sin A^2$ or $\sin 2A$.
- Using $\tan 90^\circ$ or $\sec 90^\circ$ as a number. They are not defined.
- Cancelling $\sin$ from $\dfrac{\sin A}{\sin B}$ to get $\dfrac{A}{B}$.

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